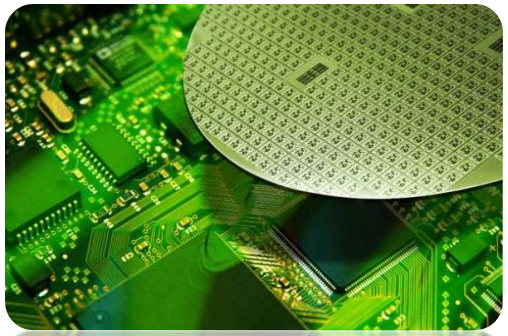
Metallization-Oxidation-Passivation





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Metallization:

- **Processes that deposit metal thin film on wafer surface.**
- **Metallizing is the general name for the technique of coating metal on the surface of objects. Metallic coatings may be decorative, protective or functional.**
- **Metallization is a general term that refers to the application of a metal coating to another metallic or non-metallic surface. *Depending on the desired result, the coating can consist of metals such as zinc, gold, aluminum or silver*.**
- **Metallization is the final step in the wafer processing sequence.**
- **Metallization is the process by which the components of IC's are interconnected**

Metallization:

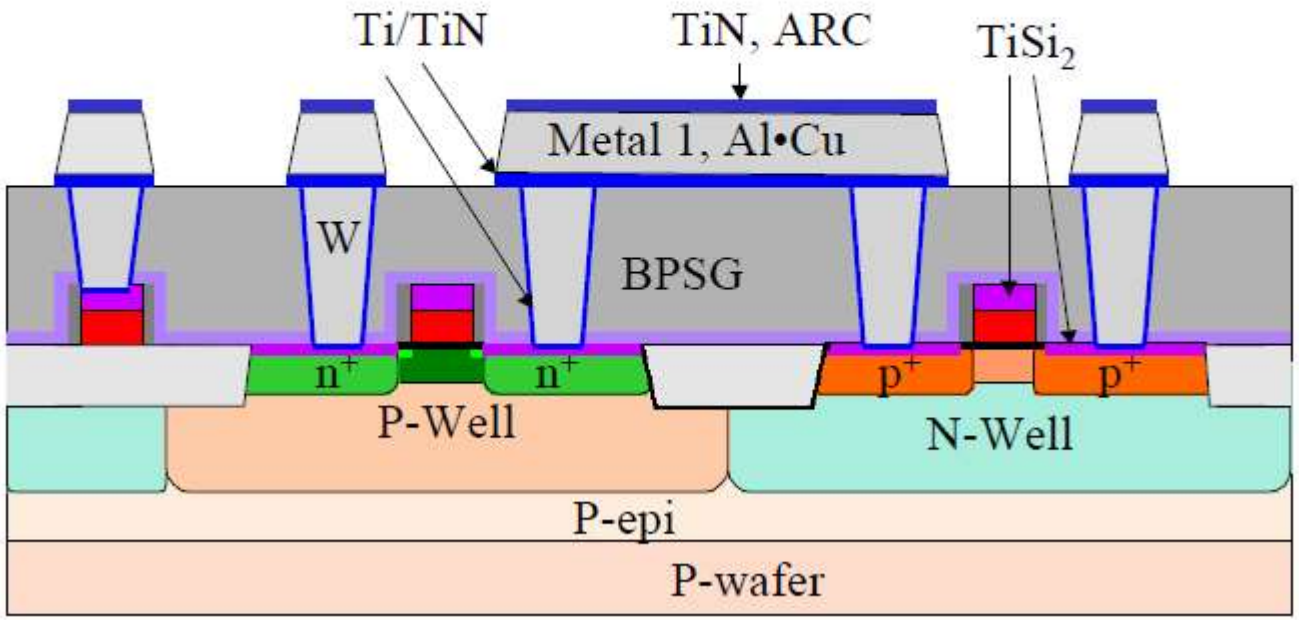
After devices have been fabricated in the silicon substrate, connections must be made to link the circuits together. This process is called metallization.

Metal layers are deposited on the wafer to form conductive pathways. The most common metals include aluminum, nickel, chromium, gold, germanium, copper, silver, titanium, tungsten, platinum, and tantalum. Selected metal alloys may also be used.

Applications:

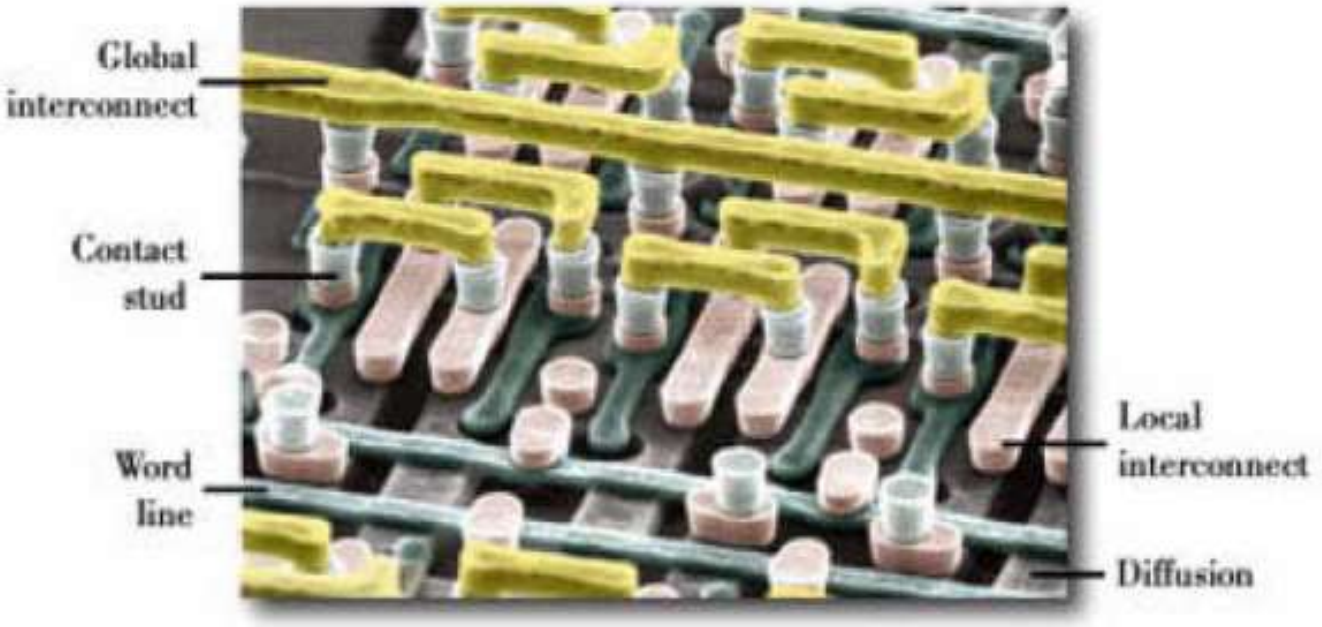
- **Interconnection**
- **Gate and electrodes**
- **Micro-mirror**
- **Fuse**

CMOS: Standard Metallization



Applications: Interconnection

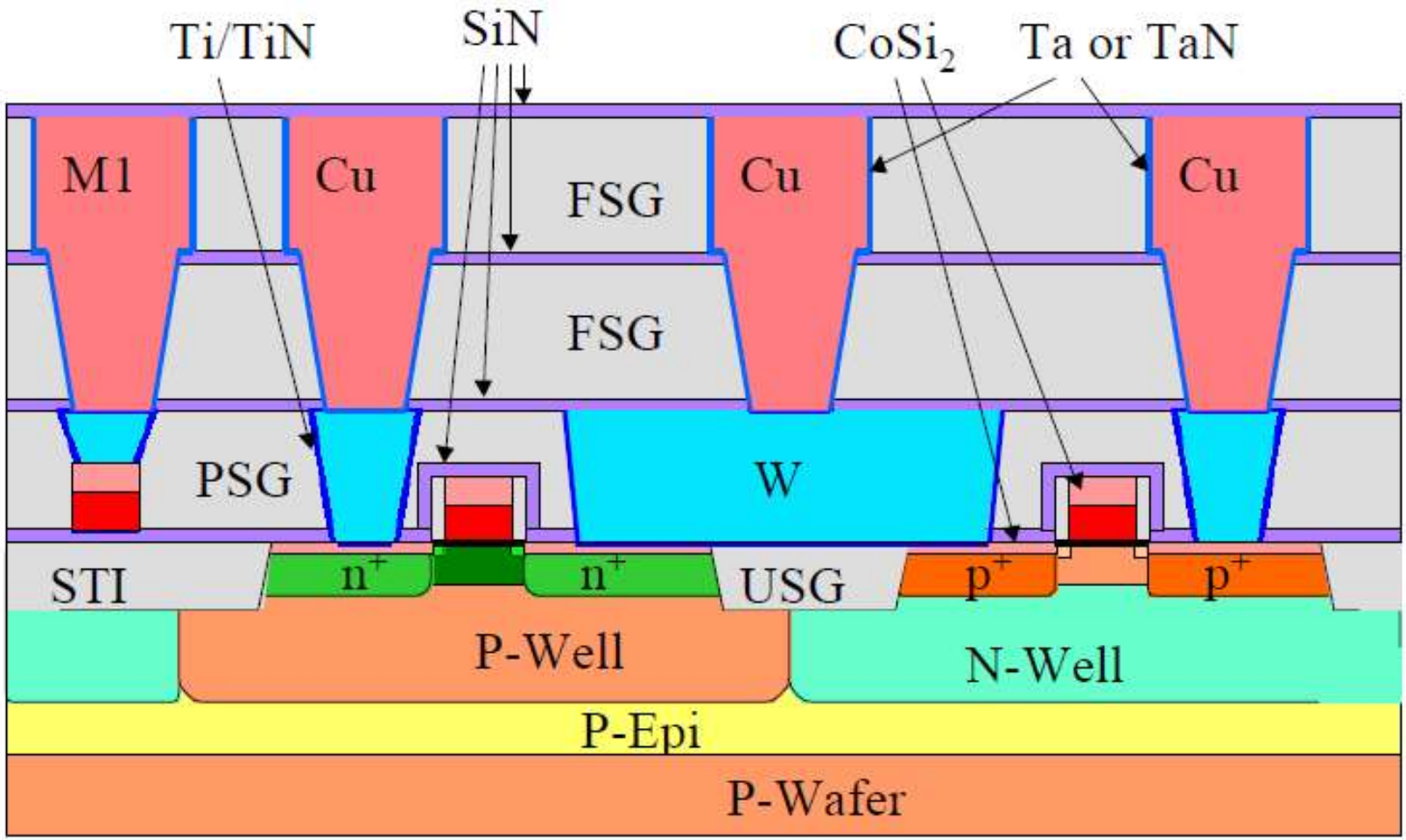
- Silver
- Gold
- Copper
- Aluminum
- Platinum



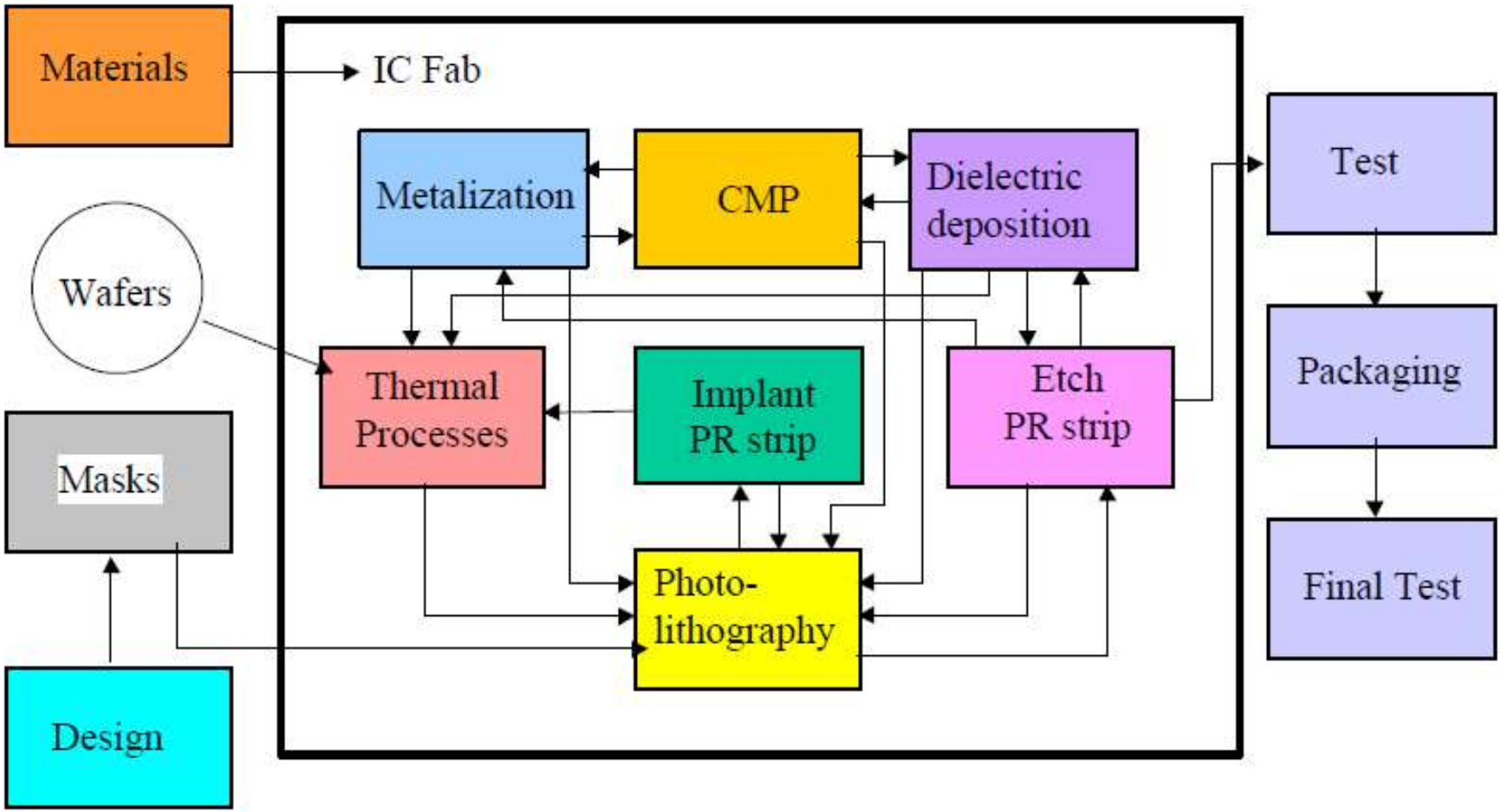
Applications: Interconnection

- Dominate the metallization processes
- Al-Cu alloy is most commonly used
- W plug, technology of 80s and 90s
- Ti, welding layer
- TiN, barrier, adhesion and ARC layers
- The future is --- Cu!

Copper Metallization



Wafer Process Flow



Conducting Thin Films:

- **Polysilicon:** Gates and local interconnections
  Replaced aluminum since mid-1970s
  High-temperature stability,
  Heavily doped

- **Silicides:** Much lower resistivity than polysilicon,
  TiSi2, WSi2, and CoSi2 are commonly used

- **Aluminum alloy:** Most commonly used metal
  The fourth best conducting metal, Silver 1.6 μΩ.cm,
  Copper 1.7 μΩ.cm, Gold silver 2.2 μΩ.cm, Aluminum 2.65 μΩ.cm
  It was used for gate before mid-1970

Aluminum-Silicon Alloy: Al make direct contact with Si at source/drain
Si dissolves in Al and Al diffuses into Si

- **Titanium:** Applications: Silicide formation, Titanium nitridation,
  Wetting layer, Welding layer

Conducting Thin Films:

- **Titanium Nitride**: Barrier layer – prevents tungsten diffusion
  Adhesion layer – help tungsten to stick on silicon oxide surface
  Anti-reflection coating (ARC) – reduce reflection and improve photolithography resolution in metal patterning process
  – prevent hillock and control electromigration

- **Tungsten:** Metal plug in contact and via holes
  Contact holes become smaller and narrower
  PVD Al alloy: bad step coverage and void
  CVD W: excellent step coverage and gap fill
  Higher resistivity: 8.0 to 12 mW×cm compare to PVD Al alloy (2.9 to 3.3 mW×cm)
  Only used for local interconnections and plugs

Conducting Thin Films:

- **Copper:** Low resistivity (1.7 mW×cm) – lower power consumption and higher IC speed
  High electromigration resistance – better reliability
  Poor adhesion with silicon dioxide
  Highly diffusive, heavy metal contamination
  Very hard to dry etch – copper-halogen have very low volatility

- **Tantalum:** Barrier layer, Prevent copper diffusion, Sputtering deposition

- **Cobalt:** Mainly used for cobalt silicide ($CoSi_2$).
  Normally deposited with a sputtering process

Metal Thin Film Measurements:

- Thickness.
- Stress
- Reflectivity
- Sheet resistance

Requirements on metallization:

The metallization realizes contacts to the doped regions in semiconductor manufacturing with conducting paths. From here the connectors are lead to the edge of the microchip to finally connect it to the package or for testing reasons.

Following requirements are essential for metallizations in integrated micro electronic devices:

- **excellent adhesion on silicon oxides (insulators between metallization layers)**
- **high ampacity, low resistance**
- **low contact resistance between metal and semiconductor**
- **simple processes to deposit the metallization layers**
- **low susceptibility to corrosion for long life times**
- **excellent contacting with wire bonds**
- **possibility to integrate stacked layers**
- **high purity of the metal**

Aluminum meets many of this requirements and therefore has been the material of choice for many years. However, since structures are getting smaller and smaller, aluminum can't fulfill the requirements any longer. For this reason, copper will replace aluminum in the future.

Metal Deposition:

Metallization is often accomplished with a vacuum deposition technique. The most common deposition processes include filament evaporation, electron-beam evaporation, flash evaporation, induction evaporation, and sputtering.

Filament Evaporation: Filament evaporation, also called resistive evaporation, is the simplest method. This process is usually carried out in a bell jar, in which a filament is heated by thermal resistance. As the temperature rises, the metal to be deposited is melted and wets the filament. The current through the filament is increased further, until the metal vaporizes. The metal vapor then condenses on the cooler surface of the semiconductor wafers, forming the desired metal layer.

Electron-Beam Evaporation: Electron-beam evaporation, frequently called "e-beam," uses a focused beam of electrons to heat the metal for deposition. The metal is kept in a water-cooled crucible and exposed to the electron beam, causing it to vaporize and condense on the wafers.

Metal Deposition:

Flash Evaporation: In flash evaporation, a ceramic bar is heated by thermal resistance. Wire is continuously fed from a spool until it contacts the heated bar. Upon contact, the metal evaporates and is deposited on the substrate.

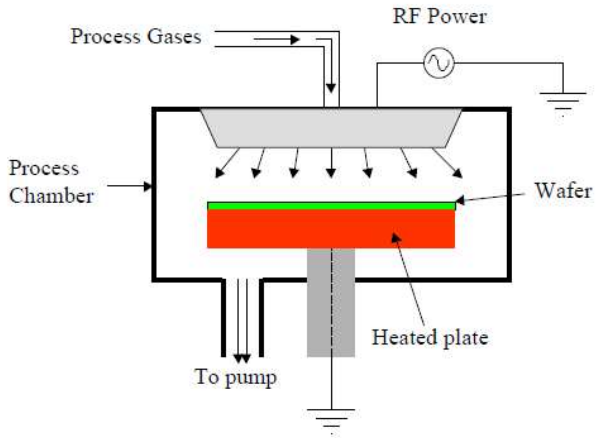
Induction Evaporation: Induction evaporation uses radio-frequency radiation to evaporate the metal in a crucible. The metal is then deposited as with other methods.

Sputtering: To perform sputtering, ions of an inert carrier gas (such as argon) are introduced into a low-pressure or partial-vacuum atmosphere. An electric field is used to ionize the atoms and draw them to one place in the chamber called the target. The target is comprised of the metal used for deposition. When the ions strike the target, they dislodge, or sputter, these metal atoms. The dislodged atoms are then deposited in a thin film on the silicon substrate facing the target. Sputtering can be done using both direct current and radiofrequency voltages and can be used to deposit almost any material.

Metal CVD

- Widely used to deposit metal
- Good step coverage and gap fill capability
  - can fill tiny contact holes to make connections between metal layers.
- Poorer quality and higher resistivity than PVD metal thin films.
  - Used for plugs and local interconnections
  - Not applied for global interconnections

Metal CVD Chamber



Physical Vapor Deposition (PVD):

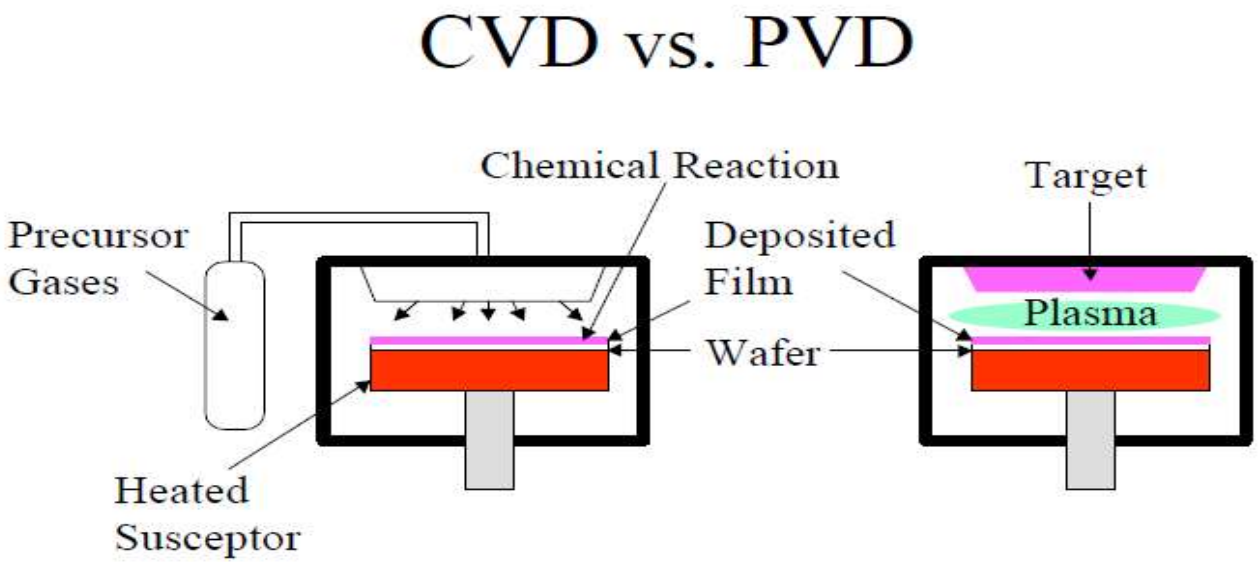
- Vaporizing solid materials
- Heating or sputtering
- Condensing vapor on the substrate surface
- Very important part of metallization

PVD vs. CVD

- PVD Start with P
- CVD Start with C

PVD vs. CVD: Sources

- PVD Solid materials
- CVD Gases or vapors



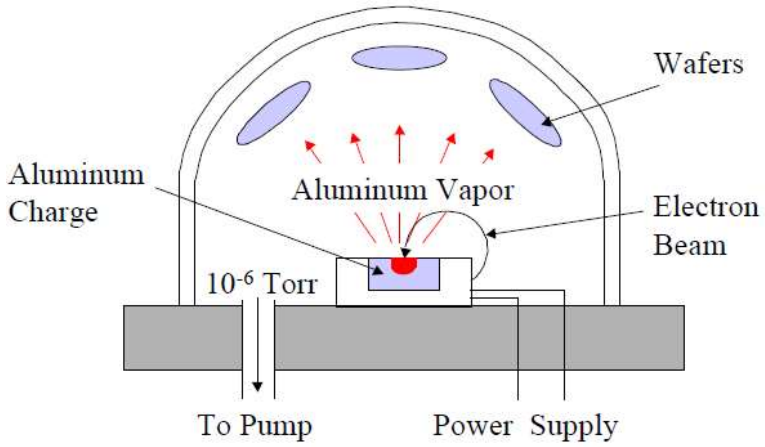
PVD Methods:

- Evaporation
- Sputtering

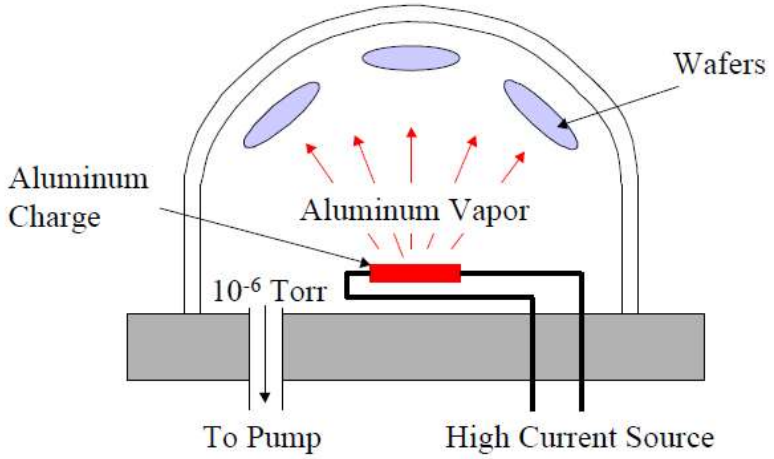
PVD Methods: Evaporation

- Filaments
- Flash hot plate
- Electron beam

Electron Beam Evaporator



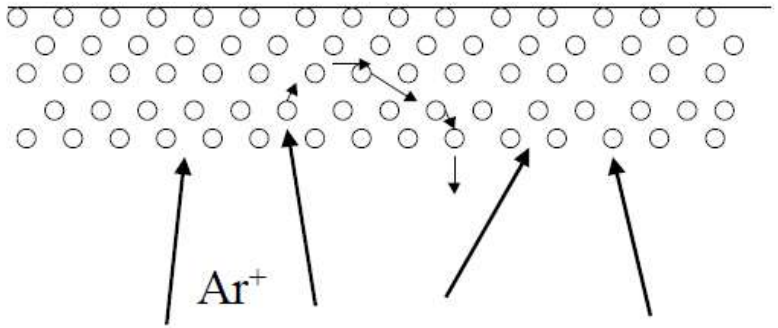
Thermal Evaporator



PVD Methods: Sputtering

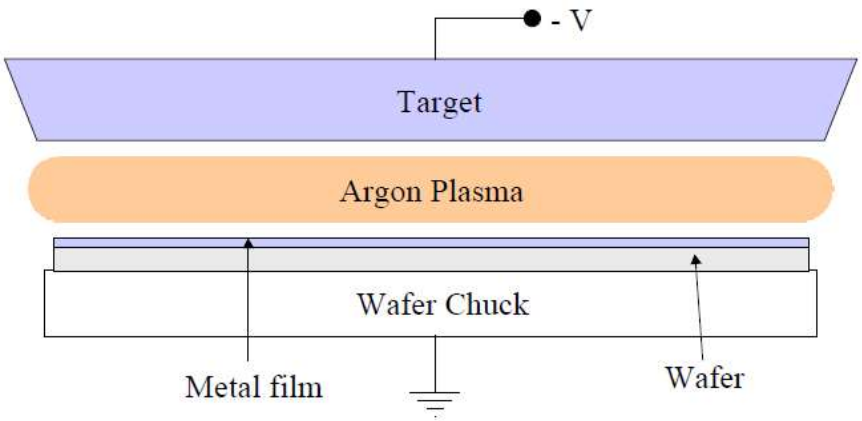
- DC Diode
- RF Diode
- Magnetron

Sputtering

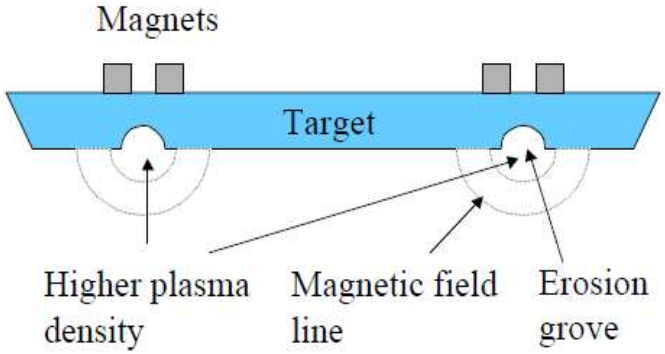


Momentum transfer will dislodge surface atoms off

DC Diode Sputtering



Schematic of Magnetron Sputtering



Oxidation

Oxidation: Overview

- **Oxidation** is a process by which a metal or semiconductor is converted to an oxide.
- **Application:** Oxides in semiconductor industry are used for multiple reasons:
  - isolation (interlayer dielectric, ...)
  - scatter oxide (ion implantation)
  - adaption layers (locos technology, ...)
  - planarization (edge rounding, ...)
  - mask layer (diffusion, ...)
  - alignment marks (lithography)
  - cover layer (to prevent corrosion, ...)

Oxidation: Overview

- In combination with silicon, oxide appears as silicon dioxide SiO_2.
- Oxidation is a process which converts silicon on the wafer into silicon dioxide.

- The chemical reaction of silicon and oxygen already starts at room temperature but stops after a very thin native oxide film.
- It can be deposited in very thin, electric stable, and uniform layers.

- Silicon dioxide layers are used as high-quality insulators or masks for ion implantation. The ability of silicon to form high quality silicon dioxide is an important reason, why silicon is still the dominating material in IC fabrication.

Oxidation on Silicon:

- Although oxidation of many materials plays a role in technology, the main oxidation reaction which will be addressed is the conversion of parts of a silicon semiconductor wafer into silicon dioxide (SiO_2).

- The chemical reaction between oxygen and silicon to generate SiO_2 is usually driven by a high-heat environment; however, even at room temperature, a shallow layer of native oxide, approximately 1nm thick can form in an air environment.

- In order to grow thicker oxides in a controlled environment, several methods can be implemented.

Oxidation on Silicon:

- Here we will consider only silicon-based (Si) technologies. Although other compound materials in groups III through V, such as gallium arsenide (GaAs) and aluminum gallium nitride (AlGaN), are also used to implement VLSI chips, silicon is still the most popular material, with excellent cost–performance trade-off.

- Recent development in SiGe and strained-silicon technologies will further strengthen the position of Si-based fabrication processes in the microelectronic industry for many more years to come.

Oxidation on Silicon

- Oxidation can be described as the production of SiO_2 on Si Substrate.
- Silicon is an abundant element and occurs naturally in the form of sand.
- It can be refined using well-established purification and crystal growth techniques.
- Silicon can be easily oxidized to form an excellent insulator, SiO_2 (glass).
- SiO_2 plays an important role in IC technology because no other semiconductor material has a native oxide which is able to achieve all the properties of SiO_2.

Functions of SiO_2

The function of a layer of silicon dioxide (SiO2) on a chip is multipurpose. Some of the important uses of SiO2 are as follows:

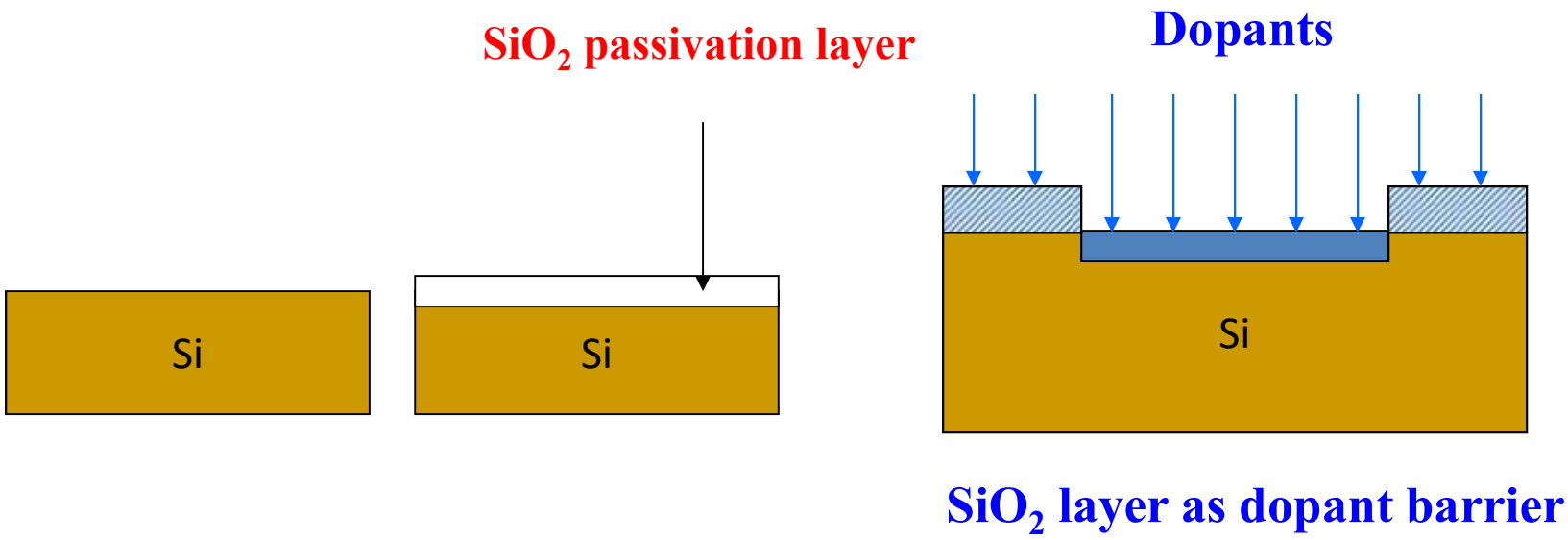
- This native oxide has a dielectric strength of approximately 10^7 V/cm and a dielectric constant of about 3.9. Thus, it is very much useful for the construction of capacitors and MOSFETs.
- In Bipolar and MOS transistors, It is used to isolate one device from another.
- It protects the junction from moisture and other atmospheric contaminants.

Functions of SiO_2

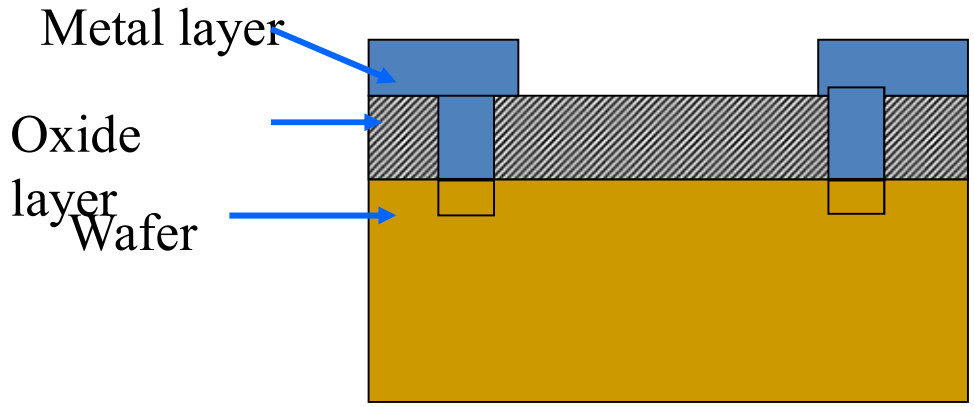
- It is used for surface passivation which is nothing but creating protective SiO2 layer on the wafer surface.
- It acts as a barrier or mask against the diffusion or the implantation of impurity dopant in substrate. (allows the introduction of dopants into the silicon only in regions that are not covered with SiO2).
- This masking property allows the electrical properties of the silicon to be altered in predefined areas
- Different active and passive elements can be built on the same piece of substrate with the help of SiO2.

Functions of SiO_2

- Using as a mask in film deposition (doping barrier)
- Protection of wafer surface against wear and scratch (Surface passivation)

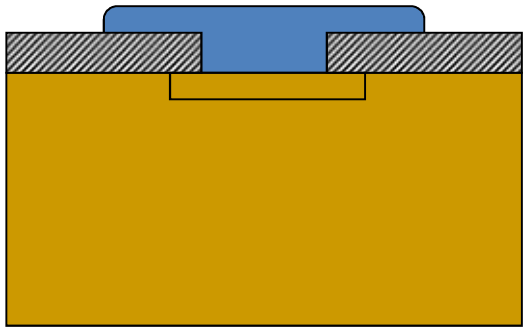


Functions of SiO_2

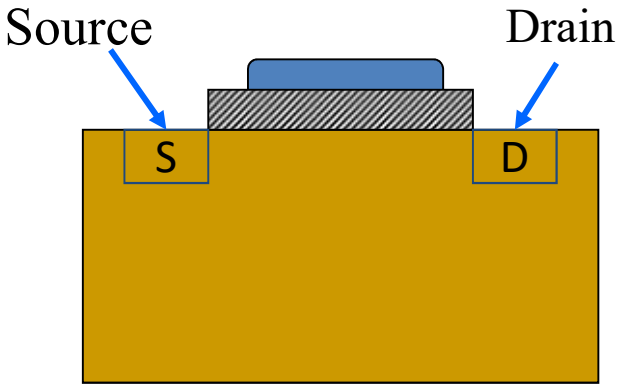


Dielectric use of SiO_2 layer

- Isolation of devices from others (surface dielectric)
- As a part in the MOSFET (Device dielectric)
- Grow thin layer SiO_2 in the gate region



Field Oxide



MOS gate

- Oxide is also used as dielectric layer in capacitors between Si wafer and conduction layer.

Functions of SiO_2

Most applications of semiconductor are dependent on their oxide thicknesses

Silicon dioxide thickness, Å	Applications
60-100	Tunneling gates
150-500	Gates oxides, capacitor dielectrics
200-500	LOCOS pad oxide
2000-5000	Masking oxides, surface passivation
3000-10000	Field oxides

VLSI Requirements for SiO_2

- **MOS VLSI technology requires silicon dioxide thickness in the 50 to 500 A range in a repeatable manner.**
- **The growth of thin oxide must be slow enough to obtain uniformity and reproducibility.**

Techniques of Oxidation

There are different techniques for the formation of SiO_2

- Thermal oxidation (Basic process used in IC fabrication)
- Wet Anodization
- Vapour phase oxidation (Also known as CVD)
- Plasma oxidation

- For thin oxide>>>>>

>Dry oxidation >Dry oxidation with HCl >Sequential oxidations using different temperatures and ambient >wet oxidation >Reduced pressure techniques and >High pressure/low temperature oxidation.

----------Ultra-thin oxide (<50 A) have been produced using hot nitric acid, boiling water, and air at room temperatures……

Thermal Oxidation

- In VLSI, thermal oxidation is a way to produce a thin layer of oxide on the surface of a wafer.
- The rate of oxide growth is often predicted by the Deal-Grove model.
- Thermal oxidation may be applied to different materials, but most commonly involves the oxidation of silicon substrates to produce silicon dioxide.

Basics of Thermal Oxidation

- In thermal oxidation, silicon reacts with oxygen to form silicon dioxide (SiO_2).
- To speed up the chemical reaction, it is necessary to carry out the oxidation at high temperatures (e.g., 1000–1200°C) and inside ultraclean furnaces.
- The oxygen used in the reaction can be introduced either as a high-purity gas (referred to as a "dry oxidation") or as steam (forming a "wet oxidation").
- The selection of oxidation technique to be used depends on oxide properties and the thickness of the oxide layer required.

Dry Oxidation

- Dry oxidation uses oxygen as shown by the chemical equation : $Si + O_2 \rightarrow SiO_2$.
- A silicon atom directly reacts with an oxygen molecule to produce one molecule of silicon dioxide.

- This type of oxidation is best for thin oxide layers with a low charge at the interface.
- Dry oxidation is also the preferred process when contamination by sodium atoms is a concern.

Characteristic of the dry oxidation:

- Slow growth of oxide
- High density
- High breakdown voltage

Wet Oxidation

- Wet oxidation is based on the chemical equation of : $Si + 2H_2O \rightarrow SiO_2 + 2H_2$
- Water in the form of steam reacts with silicon to produce silicon dioxide and hydrogen gas. (the oxygen is led through a bubbler vessel filled with heated water of about 95 °C).

- This process is used to produce thick oxide layers with relatively low temperatures and high pressure.
- This process is done by 900 to 1000°C.

Characteristics of wet thermal oxidation:

- Fast growth even on low temperatures
- Less quality than dry oxides

Growth Rate of Dry and Wet Oxidation

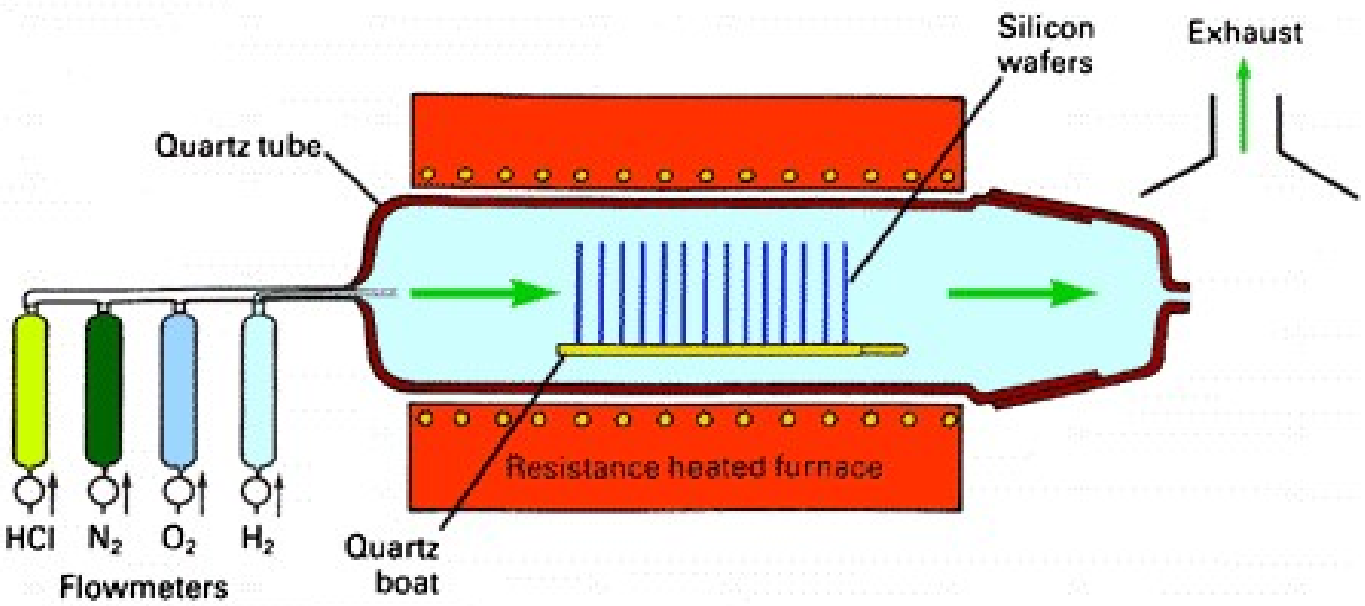
Comparison of the growth rate of wet and dry oxidation

Temperature	Dry oxidation	Wet oxidation
900°C	19 nm/h	100 nm/h
1000°C	50 nm/h	400 nm/h
1100°C	120 nm/h	630 nm/h

In general, wet oxidation has a faster growth rate, but dry oxidation gives better electrical characteristics

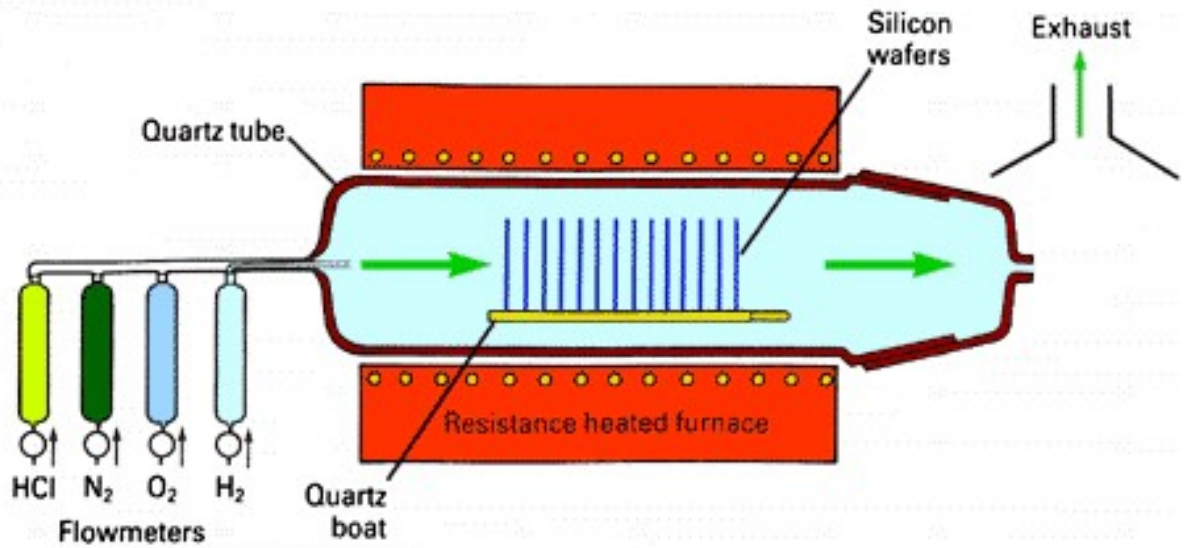
Thermal Oxidation Process

- The thermal oxidation of silicon begins by placing the silicon wafers in a quartz rack, commonly known as a boat, which is heated in a quartz thermal oxidation furnace.
- The furnaces consist of a quartz tube in which the wafers are placed on a carrier made of quartz glass.
- For heating there are several heating zones and for chemical supply multiple pipes.



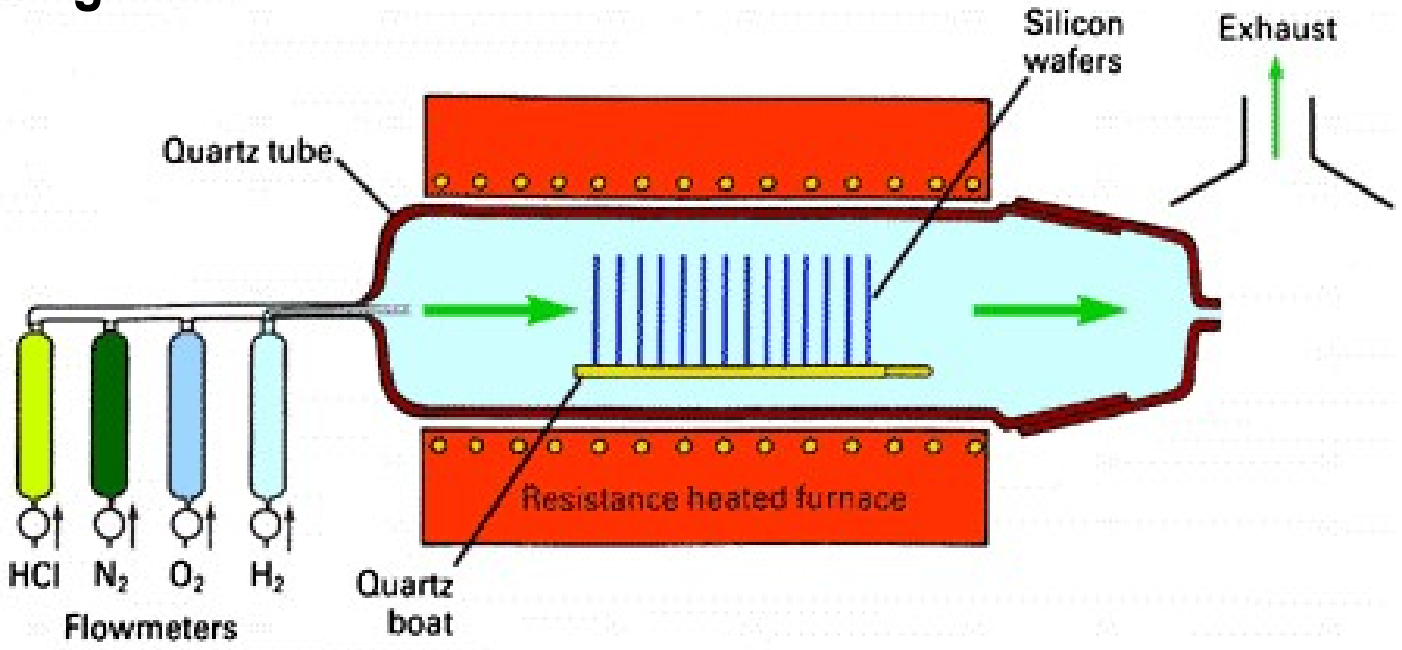
Thermal Oxidation Process

- The temperature in the furnace may be between 950 and 1250°C under standard pressure.
- Quartz glass has a very high melting point (above 1500°C) and thus is applicable for high temperature processes.
- A control system is needed to keep the wafers within about 19°C of the desired temperature.
- Oxygen or steam is introduced into the thermal oxidation furnace, depending on the type of oxidation being performed.



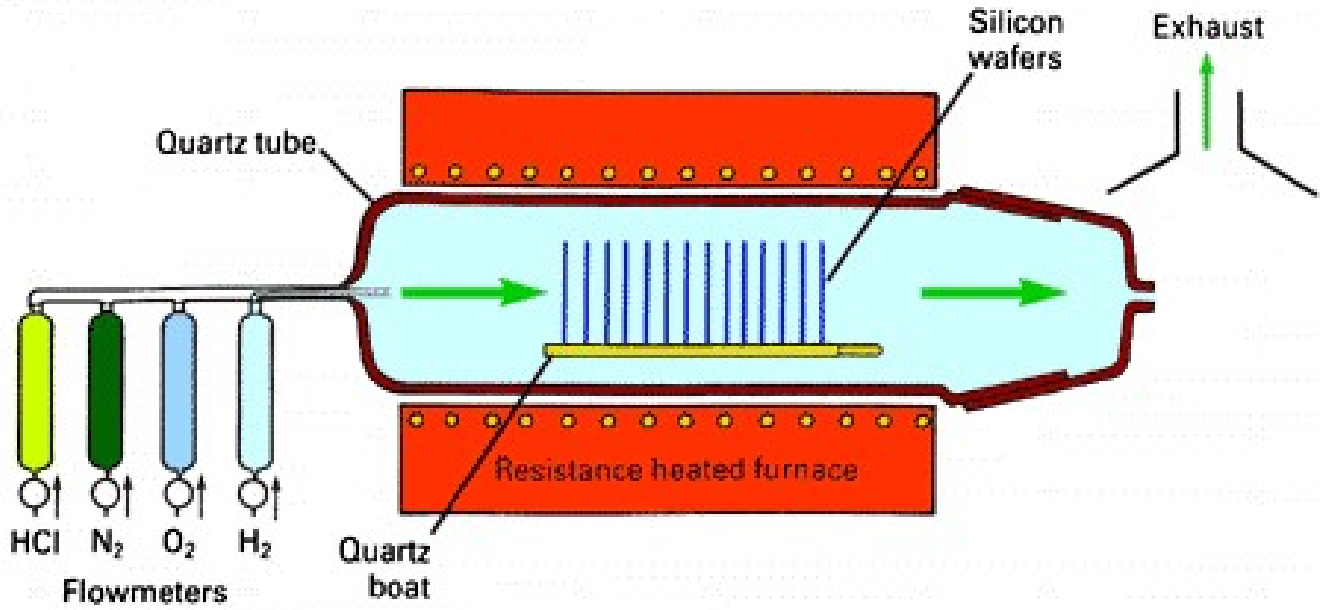
Thermal Oxidation Process

- Oxygen from these gases then diffuses from the surface of the substrate through the oxide layer to the silicon layer.
- The oxygen is led to the wafers in gaseous state and reacts at the wafer surface to form silicon dioxide.
- The composition and depth of the oxidation layer may be precisely controlled by parameters such as time, temperature, pressure and gas concentration.



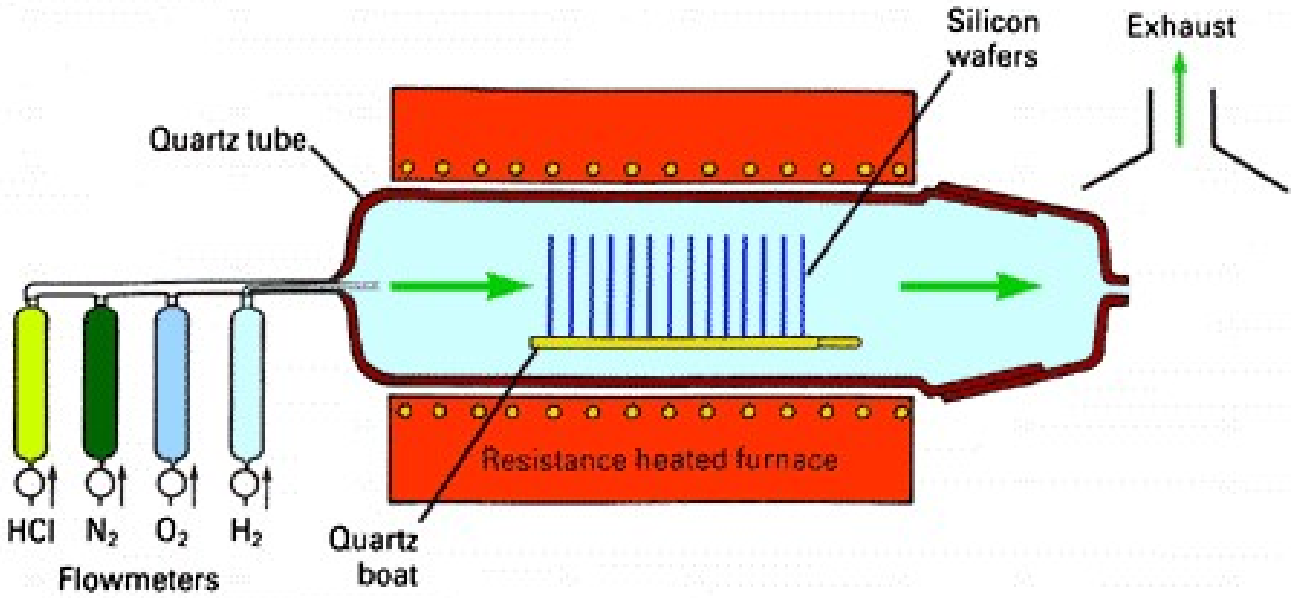
Thermal Oxidation Process

- A high temperature increases the oxidation rate, but it also increases the impurities and movement of the junction between the silicon and oxide layers.
- These characteristics are particularly undesirable when the oxidation process requires multiple steps, as is the case with complex ICs.
- A lower temperature produces an oxide layer of higher quality, but also increases the growth time.



Thermal Oxidation Process

- The typical solution to this problem is to heat the wafers at a relatively low temperature and high pressure to reduce the growth time.
- An increase of one standard atmosphere (atm) decreases the required temperature by about 20 degrees Celsius, assuming all other factors are equal.
- Industrial applications of thermal oxidation use up to 25 atm of pressure with a temperature between 700 and 900°C.



Thermal Oxidation Process

- The oxide growth rate is initially very fast but slows down as oxygen must diffuse through a thicker oxide layer to reach the silicon substrate.
- Almost 46 percent of the oxide layer penetrates the original substrate after oxidation is complete, leaving 54 percent of the oxide layer on top of the substrate.

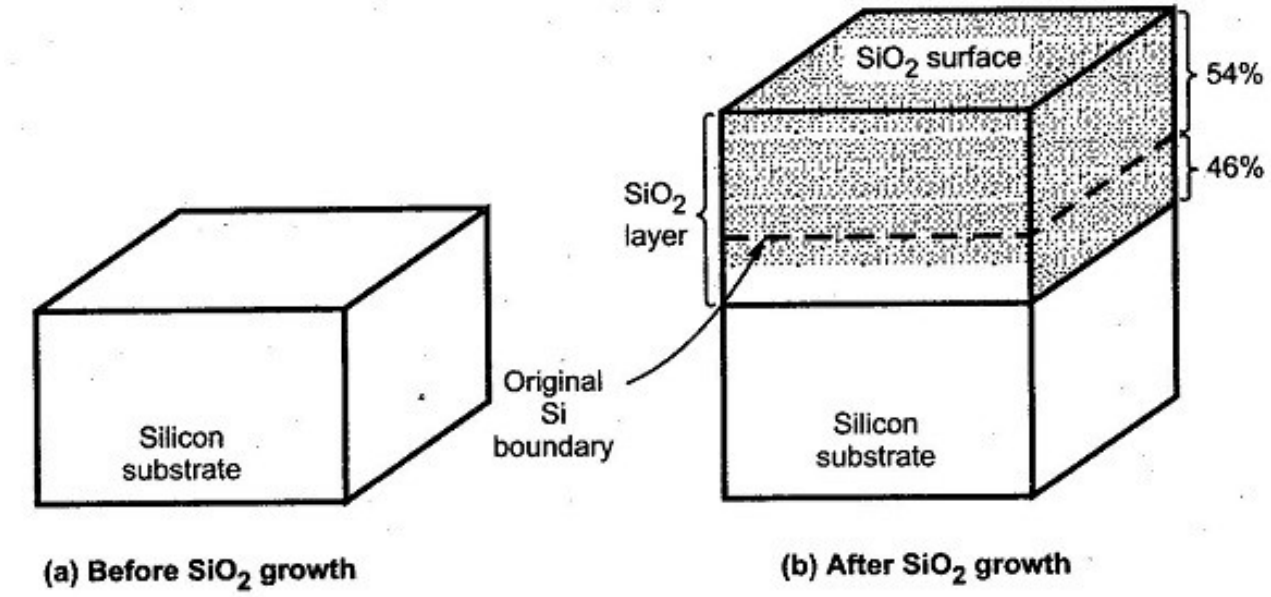


Fig. 1.4 Thermal Oxidation

Thickness Measurement

- Silicon dioxide is a transparent film, and the silicon surface is highly reflective.
- If white light is shone on an oxidized wafer, constructive and destructive interference will cause certain colors to be reflected. The wavelengths of the reflected light depend on the thickness of the oxide layer. In fact, by categorizing the color of the wafer surface, one can deduce the thickness of the oxide layer.
- The same principle is used by more sophisticated optical inferometers to measure film thickness.

Thickness Measurement

The appearance of color is due to the constructive and destructive interference of light. The wavelengths of light in SiO_2 which undergo destructive interference is given by:

$$\lambda_k = \frac{5.84t}{2k+1}$$

Where:

λ = wavelength
t = oxide thickness
k = 0, 1, 2, ... (and which give solutions in the visible light spectrum)

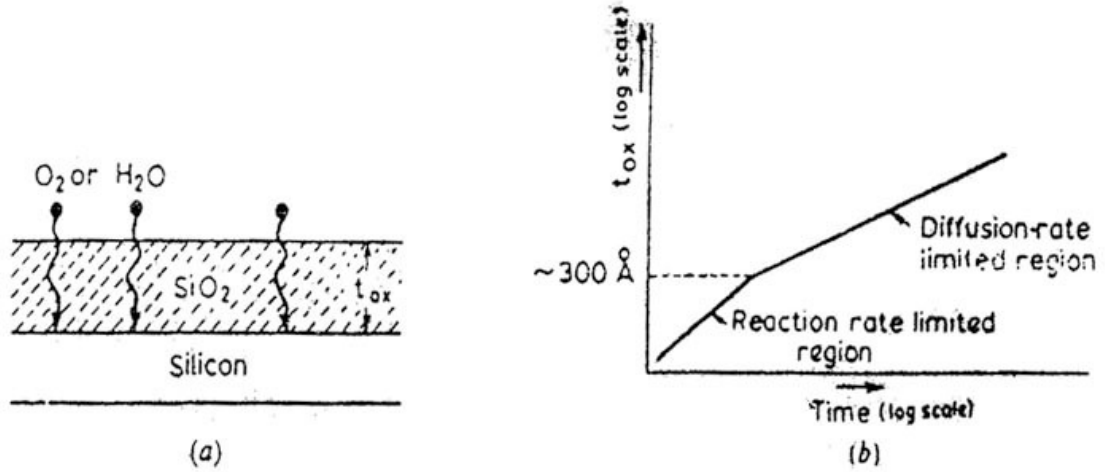
The color of light seen is the eyes response to the amplitude of the wavelengths remaining.

Film Thickness (Å)	Color of Film (those shown are only indicative)
500	tan
700	brown
1000	dark violet to red violet
1200	royal blue
1500	light blue to metallic blue
1700	metallic to very light yellow-green
2000	light gold or yellow - slightly metallic
2200	gold with slight yellow-orange
2500	orange to melon
2700	red-violet
3000	blue to violet-blue
3100	blue
3200	blue to blue-green
3400	light green
3500	green to yellow-green
3600	yellow-green
3700	green-yellow
3900	yellow
4100	light orange
4200	carnation pink
4400	violet-red
4600	red-violet
4700	violet
4800	blue-violet
4900	blue
5000	blue-green
5200	green
5400	yellow-green
5600	green-yellow

Growth Rate of Silicon Oxide Layer

- The initial growth of the oxide is limited by the rate at which the chemical reaction takes place.

- After the first 100A to 300A of oxide has been produced, the growth rate of the oxide layer will be limited principally by the rate of diffusion of the oxidant (O_2 or H_2O) through the oxide layer, as shown in the figures given below.



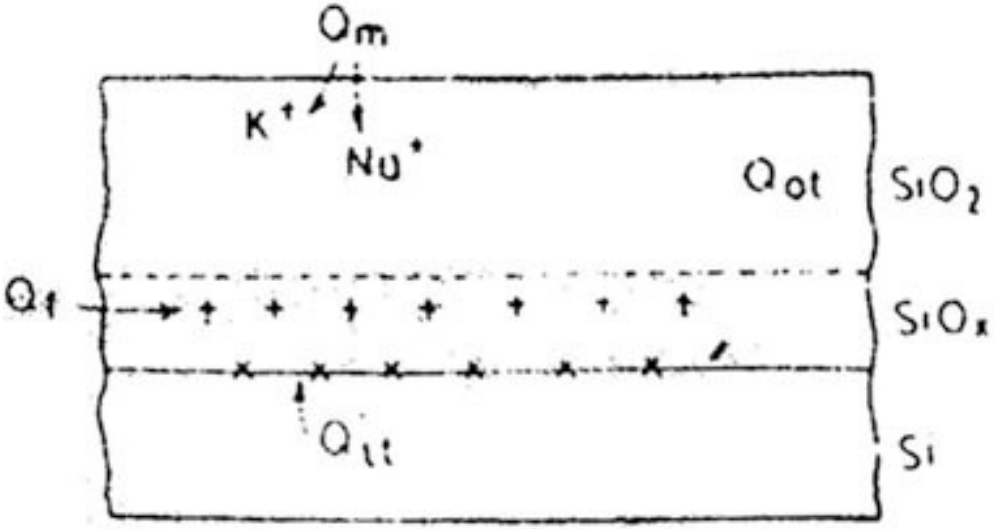
(*a*) Oxidation of silicon

(*b*) Oxide-thickness (t_{ox}) versus time for thermal oxidation of silicon.

The rate of diffusion of O_2 or H_2O through the oxide layer will be inversely proportional to the thickness

Oxide Charges

- The thermally oxidized silicon consists different types of charges. There exists a transition region at the silicon/silicon-dioxide interface.
- A charge at the interface can induce a charge of the opposite polarity in the underlying silicon, thereby affecting the ideal characteristics of the MOS device.



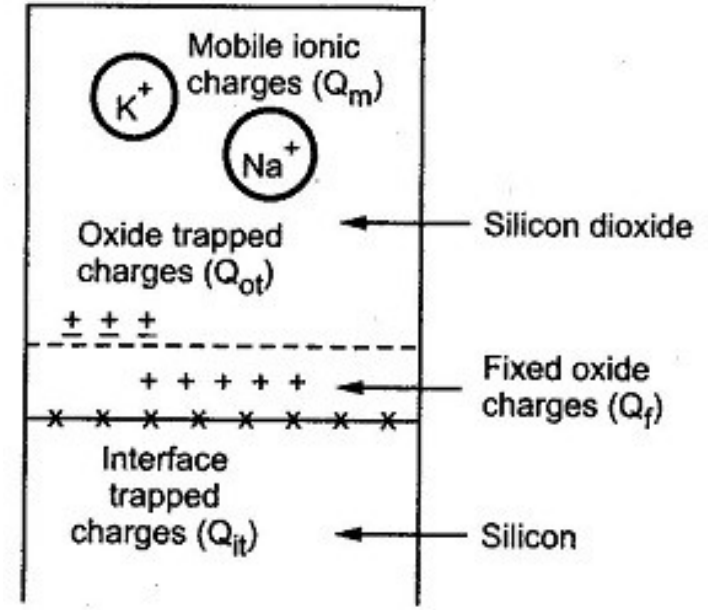


Fig. 1.5 Charges in oxidized silicon

Oxide Charges

- Thermal oxidation imparts electrical charges on the oxide layer, whereas the silicon substrate should be electrically neutral.

- Electrical charges in a silicon wafer may be categorized into three types, consisting of :

  - Fixed oxide charge (Q_f),
  - Mobile ionic charge (Q_m) and
  - Oxide trapped charge (Q_t).

Oxide Charges

- The fixed oxide charge Qf is usually positive. This can not be charged or discharged. It is not allowed beyond 30A° of Si — SiO2 interface.

- The mobile ionic charge Qm is positive most of the times. The alkali ions such as sodium, potassium are referred as mobile ion charge. But the negative ions and heavy metals are also referred as ionic charges.

- The oxide trapped charge Qot is either positive or negative. It arises from either holes or electrons trapped in the oxide. The main cause of Qot is the defects in the oxides. Also due to avalanche injection, ionizing radiations and high currents in the oxides, Qot may be induced.

Oxide Charges

- In addition to these charges, there is an interface trapped charge Qit. result from several sources including structural defects, metallic impurities, or bond breaking processes.

All these charges are calculated using capacitance voltage (C–V) analysis technique

Oxidation by Vapor Deposition

In deposition processes, oxygen and silicon are added in gaseous states. There are two important processes for oxidation by vapor deposition:

- The Silane Pyrolysis and
- The TEOS Deposition

Oxidation by Silane Pyrolysis

- **Pyrolysis means a cleavage of chemical compounds - in this case the gas silane (SiH4) and highly purified oxygen O2 - by heat.**

- **At about 400 °C silane reacts with oxygen to form silicon dioxide and hydrogen which is exhausted:**

$$SiH_4 + O_2 \rightarrow SiO_2 + 2H_2$$

Oxidation by TEOS Deposition

- The liquid TEOS (SiO4C8H2O) which is used in this process contains the required elements silicon and oxygen.

- Under a vacuum the liquid transforms into gas and is led into a tempered quartz tube with the wafers at about 750 °C where it is cleaved.

$$SiO_4C_8H_2O \rightarrow SiO_2 + \text{byproducts}$$

- The silicon dioxide deposits on the wafer, byproducts (like H2O in gaseous state) are exhausted.

Passivation

Passivation:

Passivation refers to a material becoming "passive," that is, less affected or corroded by the environment of future use.

Passivation involves creation of an outer layer of shield material that is applied as a microcoating, created by chemical reaction with the base material, or allowed to build from spontaneous oxidation in the air.

Surface passivation refers to a common semiconductor device fabrication process critical to modern electronics. It is the process by which a semiconductor surface is rendered inert, and does not change semiconductor properties as a result of interaction with air or other materials in contact with the surface or edge of the crystal. This is typically achieved using a form of thermal oxidation.

Passivation:

Passivation: rendering inactive; term typically refers to the process in which chemically and electrically active broken bonds at the semiconductor surface are saturated, and hence de-activated, by reaction with selected element; e.g. hydrogen passivates broken Si bonds at the surface; oxide grow on Si surface passivates it as well.

Surface passivation: termination of bonds on the semiconductor surface with elements assuring chemical stability of the surface; surface is rendered chemically "passive"; e.g., hydrogen termination of bonds on the silicon surface will prevent oxidation of the surface; oxidation of Si surface bonds will also passivate the surface.

Hydrogen termination: broken silicon bonds on the surface are saturated with hydrogen; Si-H is a stable bond preventing oxidation of silicon at room temperature

Passivation:

In a silicon semiconductor, this process allows electricity to reliably penetrate to the conducting silicon below the surface, and to overcome the surface states that prevent electricity from reaching the semiconducting layer.

Passivation is the process of treating or coating a metal in order to reduce the chemical reactivity of its surface. In stainless steel, passivation means removing the free iron from the surface of the metal using an acid solution to prevent rust.

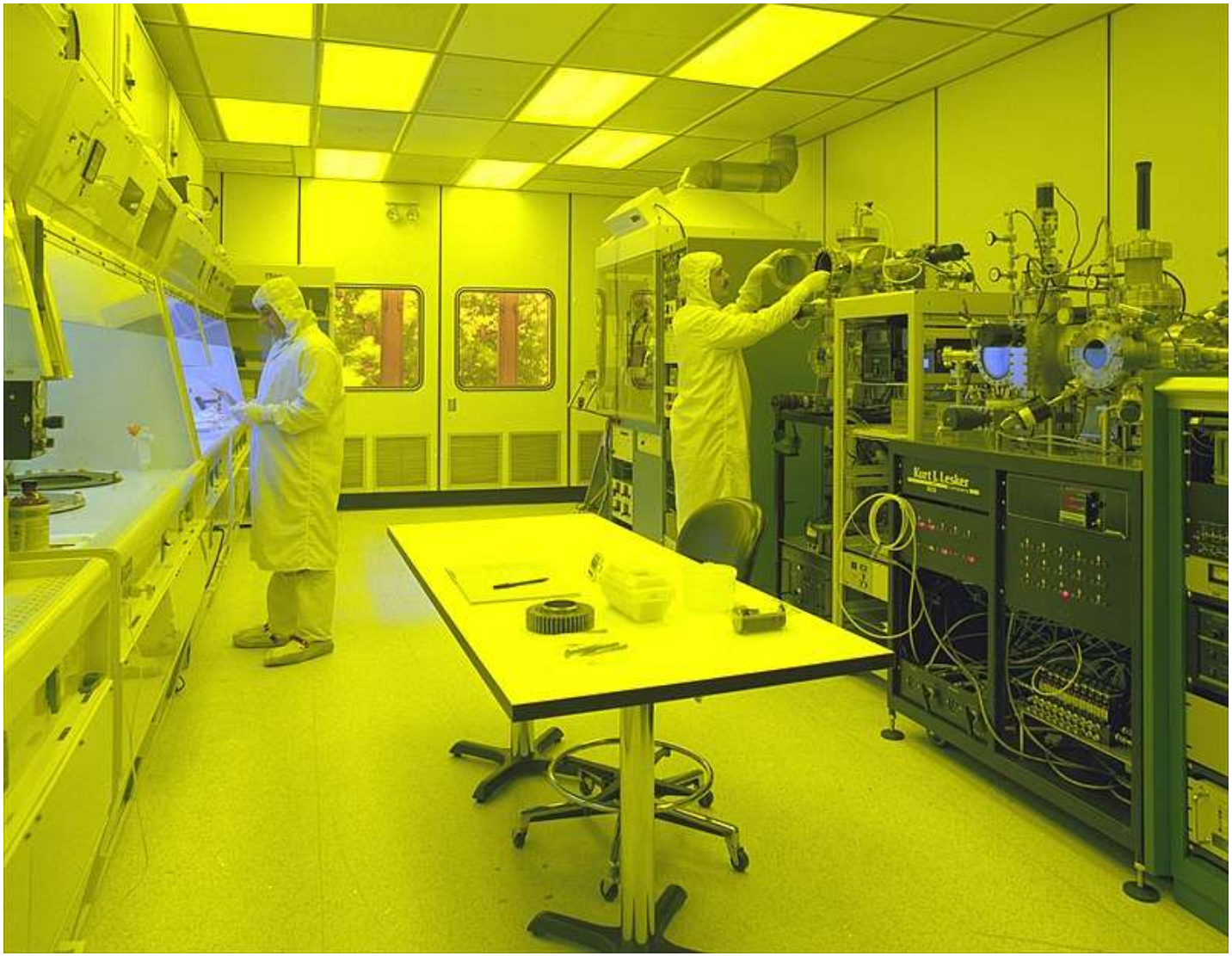
Clean Room:

- To avoid the introduction of even small quantities of contaminants (which could significantly alter the electrical properties of the silicon), it is necessary to operate it in a clean room.

- Particle filters are used to ensure that the airflow in the processing area is free from dust.

- All personnel must protect the clean-room environment by wearing special lint-free clothing that covers a person from head to toe.

Clean Room:



Clean Room:



Clean Room:

